Exercise 5
Advanced SQL
NULL FUNCTIONS

1 Show all employees with their salary. If salary is NULL, display 0. Expected columns: employee_id, name, salary_with_default

```
Select employee_id, name,
       IfNull(salary, 0) As salary_with_default
From Employees;
```

Employees

Employee_id	Name	Salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

2 Show employee names with their department. If department is NULL, show 'Not Assigned'.

```
Select name,
       Coalesce(department, 'Not Assigned') As department_name
from Employees;
```

Employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

3 Find orders with Null customer_id using ISNULL().

```
Select order_id, customer_id
From Orders
Where customer_id IS NULL;
```

Order_id	Customer_id
103	NULL

4 Show all orders. If delivery_date IS NULL, show 'Pending'.

~~Select order_id, order_ customer~~

```
Select order_id, Customer_id,
       Coalesce(delivery_date, Pending) As delivery_status
From Orders;
```

Order_id	Customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

5 Show all students and their grades. Replace NULL with 0.

```
Select Student_id, name,
       IfNull(grade, 0) As final_grade
From Students;
```

Student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

6. Count students who haven't been graded

```
Select Count(*) As not_graded_count
From Students
Where grade IS NULL
```

not_graded_count
1

7 Show product name, price and final price after discount (assume 0 if discount is NULL).

```
Select product_id, name,
       IFNULL(discount,0) As final_price
From Products
Where Price -(price * IFNULL(discount,0)/100) As finalprice;
```

product_id	name	final_price
501	Keyboard	25
502	Mouse	5
503	Monitor	100

8 Count how many customers have no email

```
Select Count(*) As missing_email_count
 From Customers
Where email IS NULL;
```

Missing_email_count
2

9 Show all customers with email. If-Null, display "No Email".

```
Select customer_id, name,
 Coalesce(email, 'No Email') As email_disply
From Customers;
```

Customer_id	name	email_display
1	Linda	No Email
2	Joseph	Joseph@mail.com
3	Nia	No Email

10 Show payment details with method replaced by 'Unknown' if NULL

```
Select payment_id, status,
 Coalesce(method, Unknown) As method_display
From Payments;
```

payment_id	method_display	Status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

11 Show items and their quantity (0 IFNULL).

```
Select Item_id, Item_name,
  IfNull(quantity, 0) As quantity_Checked.
  From Inventory;
```

Item_id	Item_name	quantity_checked.
1	Pen	0
2	Notebook	150
3	Eraser	0

12 Show employeeID and the first available value among bonus or commission

```
Select empdo emp_id,
    Coalesce(bonus, Commission, 0) As first_available_reward
From Employees_Extra;
```

emp_id	First available reward
1	300
2	100
3	0

13. Count classes that dont have a room assigned

```
Select Count (*) As no_room_count
From Classes
Where ISNULL;
```

no_room_count
2

14 Show attendance records with status. Replace NULL with 'NOT Marked'

```
Select student_id, date,
    Coalesce(status, 'Not Marked'
From Attendance ;
```

student_id	date	attendance_status ~~checked~~
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

15 Show accountID, account_type (or 'Unknown'), and balance (or 0).

```
Seled accountID,
    Coalesce (account_type,('Unknown') As type_display,
    IfNull (balance, 0) As balance_checked
From Bank_Accounts ;
```

~~Account_id~~ Account_id	account type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

16 Show all projects with a start date. If start_date is NULL, display 'TBD'.

```
Select project_id, title,
Coalesce (start_date, 'TBD') As start_disply
From Projects ;
```

Project_id	title	Start_disply
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

17 Display reviews showing comment (or 'No Comment') and rating (or 0)

```
Select review_id, product_id,
    Coalesce( comment, 'No Comment') As comment_display,
    IFNull (rating, 0) As rating_display
from Reviews ;
```

Sele

Review_id	Product_id	Comment_disply	rating_display
1	Great product		
2	No comment		
3			

ReviewID	Product_id	Comment_disply	rating_disply
1	501	Great product	4
2	502	No Comment	0
3	503	Works fine	3

18 Show the supplier contact number. Use Coalesce (phone, alt_phone, 'No contact').

```
Select supplier_id, name,
  Coalesce(phone, alt_phone, 'No Contact') As contact_number
From Suppliers ;
```

Sele. Supplier_id	name	Contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	Value Source	No Contact

19 Show all users and their preferences. Replace all NULLs with defaults:
- Theme -> 'Light'
- Language -> "English"
- Timezone -> "UTC"

```
Select user_id,
  Coalesce (theme, 'Light') As theme_set,
  Coalesce (language, 'English') As language_set,
  Coalesce (Timezone, 'UTC') As timezone_set
From User_Settings ;
```

user_id	theme_set	Language_set	timezone_set
1	light	English	UTC
2	Dark	English	UTC+1
3	Tight	English	UTC

20 Show maintenance log with:
- issue -> default to "Unknown Issue"
- technician -> default to "Not Assigned"

```
Select record_id, machine_id,
  Coalesce (issue, 'Unknown issue') As issue_log,
  Coalesce (technician, 'Not Assigned') As technician_name
From Maintenance;
```

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex

Exercise 6